MATLAB

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LABELS AND ANNOTATIONS

MATLAB Drive

Editor

experiment3adc.m | experiment4adc.m | experiment6adc.m | exp1dsp.m

/MATLAB Drive/exp1dsp.m

```
clc; clear; close all;

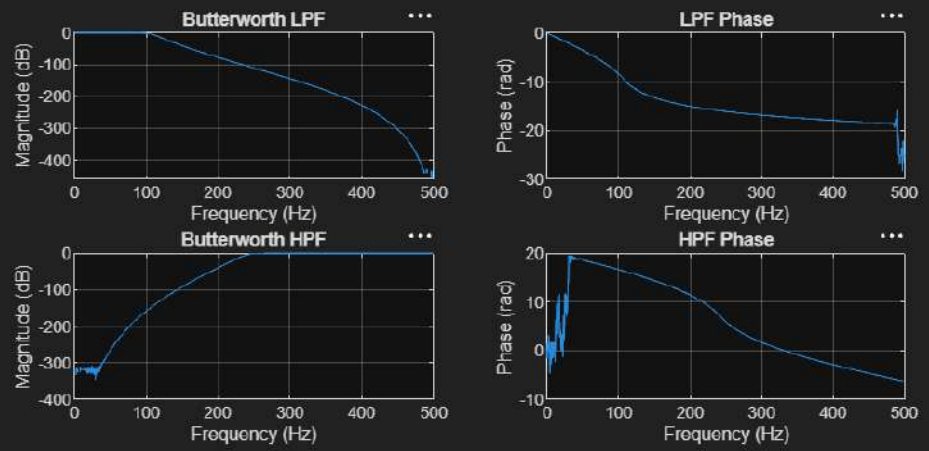
%% Parameters
Fs = 1000;
FpLP = 100;  FstLP = 150;
FstHP = 200; FpHP = 250;
Ap = 1;  Ast = 40;

%% Butterworth LPF
[NL,WL] = buttord(FpLP/(Fs/2),FstLP/(Fs/2),Ap,Ast);
[bL,aL] = butter(NL,WL,'low');
[HL,f] = freqz(bL,aL,2048,Fs);

%% Butterworth HPF
[NH,WH] = buttord(FpHP/(Fs/2),FstHP/(Fs/2),Ap,Ast);
[bH,aH] = butter(NH,WH,'high');
[HH,~] = freqz(bH,aH,2048,Fs);
```

Figures

Figure 1 | Figure 2



Command Window

```
LPF Order = 12
HPF Order = 17
LPF Numerator:
   1.0e-03 *

    0.0002    0.0027    0.0146    0.0488    0.1098    0.1757    0.2050    0.1757    0.1098    0.0488    0.0146    0.0027    0.0002

LPF Denominator:
    1.0000   -6.8844   22.4341  -45.5100   63.7725  -64.8501   48.9643  -27.6092   11.5218   -3.4662    0.7128   -0.0899    0.0053
```

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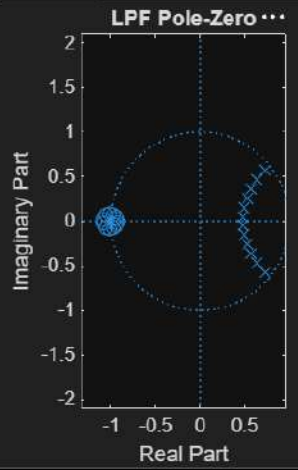
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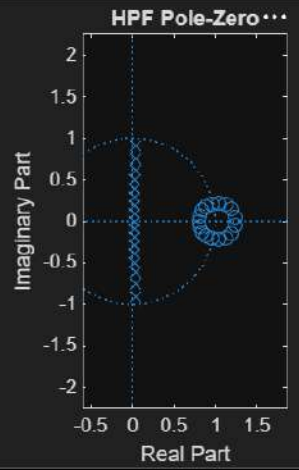
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HPF Numerator:
    0.0001   -0.0013    0.0100   -0.0502    0.1756   -0.4566    0.9132   -1.4350    1.7938   -1.7938    1.4350   -0.9132    0.4566   -0.1756    0.0502   -0.0100    0.0013   -0.0001

HPF Denominator:
    1.0000   -0.5247    2.4322   -1.0556    2.2461   -0.7954    1.0097   -0.2858    0.2357   -0.0517    0.0283   -0.0046    0.0016   -0.0002    0.0000   -0.0000    0.0000   -0.0000

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```

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